

ASSESSING THE SEVERITY OF TRAFFIC ACCIDENTS IN THE USA

Introduction

Traffic accidents have long remained a leading cause of injury and fatality in the United States. According to the Bureau of Transportation Statistics¹, the annual number of deaths due to vehicle collisions has hovered above 32,000 since the start of the 21st century. In this project, I take a unique approach to differentiate between a minor and a severe accident by focusing solely on temporal and surrounding stimuli. Two questions guided my investigation:

- Is there a particular time (day, week, season) associated with elevated accident risks?
- How do road environments and weather conditions impact accident outcomes?

I discovered that there are clear trends and patterns between the predictors above and the severity of the accident. My findings also reveal a machine learning algorithm that performs moderately well in predicting crash severity.

Dataset

My analysis uses the [US Accidents \(2016 – 2023\)](#) data from Kaggle. It compiles detailed records of 7.7 million traffic accidents across the continental United States over recent years. Of the 46 available columns, the important predictors of this study are:

- Accident timestamp
- County and state
- Road infrastructure indicators such as presence of **junction**, **bump**, and **railway**
- Weather features including **temperature**, **humidity**, and **condition**

On the other hand, the key response variable is a four-level factor called **accident severity**.

Methodology

Prior to performing any in-depth investigation, I first preprocessed the dataset. This process involves removing rows with missing values; mapping accident timestamp to its corresponding temporal features; and grouping granular weather conditions into broader categories. To ensure computational efficiency and avoid class imbalance issues, I sampled 50,000 rows of each class and consequently categorized levels 1 & 2 as **Minor** and 3 & 4 as **Severe**.

Upon cleaning the dataset, I conducted an exploratory data analysis (EDA) with visualization and statistical summaries to allow me to understand potential relationships. Building on these

1. Bureau of Transportation Statistics. (2024). *Motor vehicle safety data*.
<https://www.bts.gov/content/motor-vehicle-safety-data>

insights, I constructed two classifiers, namely **logit with bi-directional stepwise selection** (baseline) and **random forest**, to compare how one model performs against another in terms of accuracy and AUC score. In this step, I used an 80/20 train-test split, standardized the numeric features accordingly, and did cross-validation to tune the hyperparameters involved. Finally, I created a feature importance plot to highlight the most influential variables.

Findings

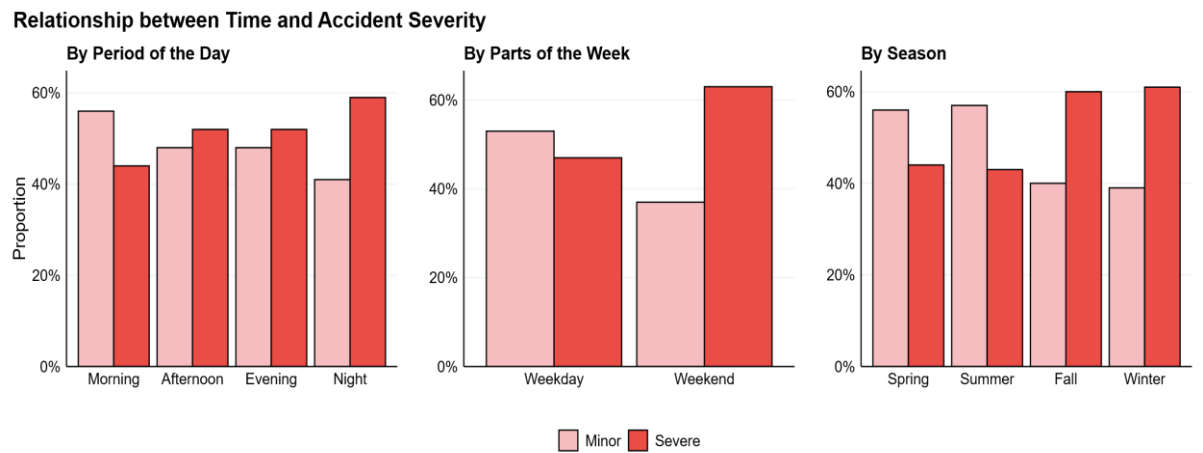


Figure 1

As part of my EDA, I examined the effects of time on crash severity. **Figure 1** shows that the seriousness of an accident does vary by time. Generally, severe crashes are more frequent during the latter parts of the day, the weekends, Fall, and Winter. These patterns align with the fact that late afternoons coincide with rush hours and reduced visibility, Saturdays/Sundays have higher traffic volumes, and colder seasons bring slippery roads.

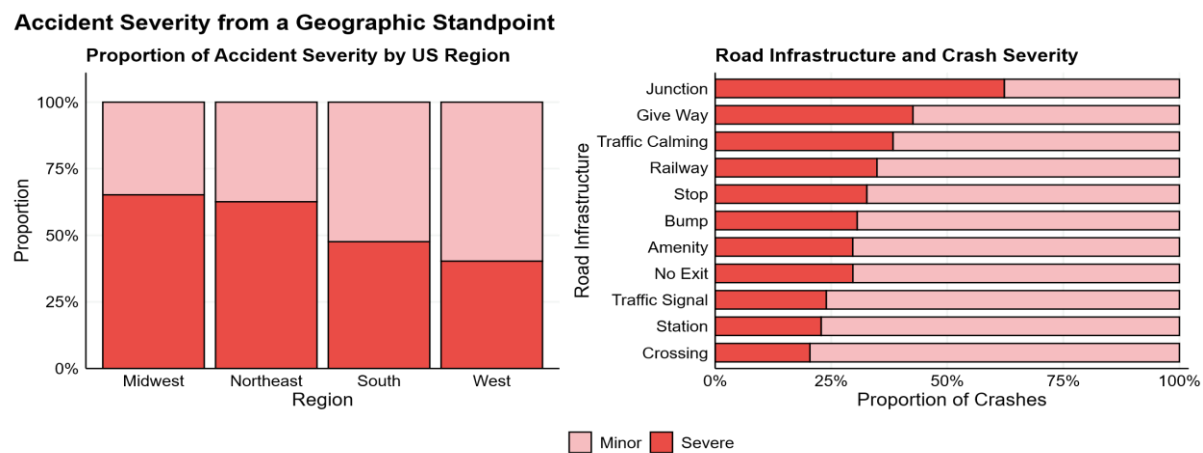


Figure 2

I also explored how geographic variables impact accident severity. It is rather alarming but unsurprising to find that the roads in Midwest and Northeast cause more severe crashes compared to the other two regions. According to the data by the Federal Highway Administration², six of the ten states with the poorest road conditions are situated in these regions. These roads make it difficult for vehicles to navigate and can cause loss of control. **Figure 2** also displays how the presence of various road infrastructure impact accident severity. Junctions top this chart, perhaps due to the multiple directions of travel or running red lights.

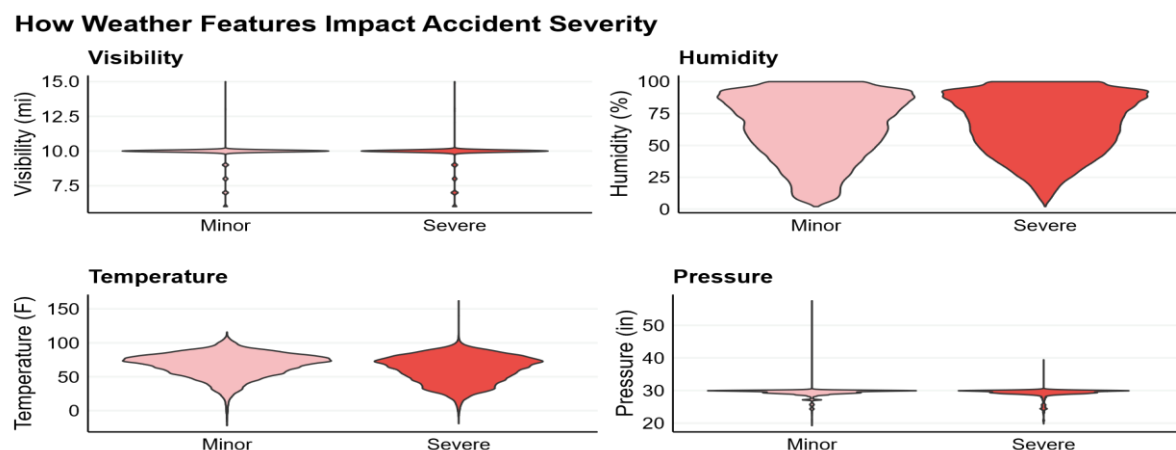


Figure 3

I then looked at the distribution of various weather features to identify any distinct patterns between minor and severe crashes. Visually, the violin plots show fairly similar distributions across visibility and humidity, with slight differences in temperature and pressure. To that end, I decided to look deeper by running individual significance tests. I discovered that these four features were highly significant ($p < 2.2e^{-16}$). Lower visibility, temperature, and pressure along with higher humidity correspond to an increased likelihood of a severe crash.

Putting it all together in two classification algorithms, I managed to garner scores that signify robust discrimination between minor and severe crashes. Using logit with bi-directional AIC-based stepwise selection as my baseline, **amenity**, **stop sign**, and **bump** were removed from the predictors. This optimization method helps prevent multicollinearity and validate other core assumptions. Nonetheless, I wanted a better-performing model; thus, I turned to random forest, an ensemble learning algorithm that can capture non-linear relationships. Leveraging the **caret** package to tune the hyperparameters, the final random forest model uses **Gini** for splits with an **mtry** value of 10 and a **minimum node size** set to 1.

2. Construction Coverage. (2024). *U.S. states with the worst roads*.

<https://constructioncoverage.com/research/states-with-the-worst-roads>

Model 1 (Logistic Regression)		Model 2 (Random Forest)	
Accuracy	AUC	Accuracy	AUC
0.63	0.72	0.69	0.76

Table 1

As a result, the random forest algorithm increases the accuracy by approximately 11% and AUC score by 6% compared to the baseline logistic regression model. I decided to proceed with the ensemble learning method to identify the top covariates of accident severity.

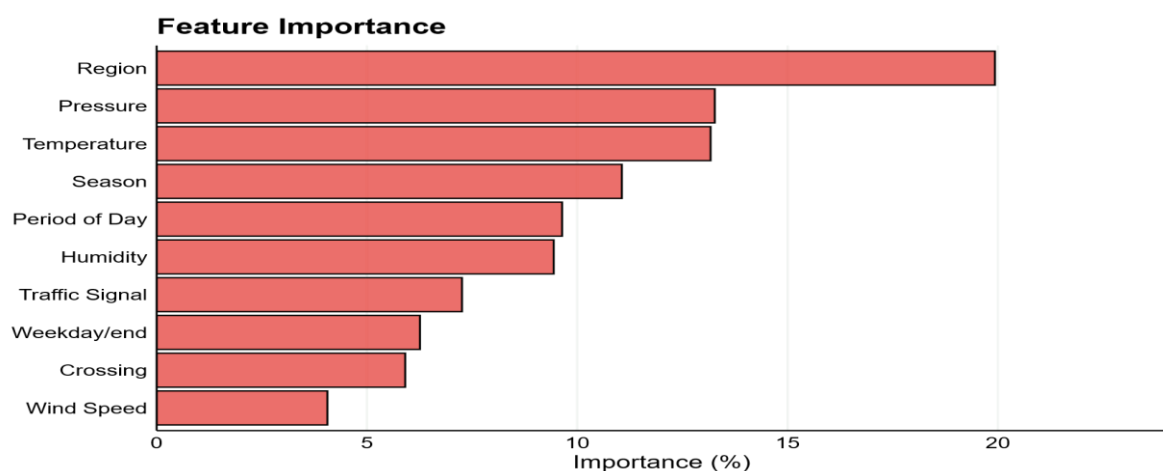


Figure 4

By plotting a graph of feature importance, the three best predictors happen to be **region**, **atmospheric pressure**, and **temperature**. Road conditions in the Northeast and Midwest regions are particularly worse compared to those in Southern and Western United States. In turn, crashes are more severe in the former areas. From a different view, Wang et al.³ mentions that air pressure can affect motor insulation in EVs and thus reduces their reliability for safe operation. Finally, an increase in temperature affects driver fatigue and reaction time, while a decrease in temperature could cause issues to a vehicle's battery power and braking system.

Implications

The significant temporal trends suggest that the government and law enforcement agencies should implement interventions that are more strategically timed. For instance, they could improve road lighting at night, most importantly at highways where the volume of vehicles are

3. Wang et al. (2022). *Impact of air pressure variations on electrical vehicle motor insulation*. <https://ietresearch.onlinelibrary.wiley.com/doi/pdf/10.1049/hve2.12346>

consistently high. Likewise, more resources need to be allocated to de-ice the roads during the colder seasons.

From a geographical standpoint, these responsible bodies must be more proactive in carrying out resurfacing/pothole-repairing projects at the affected sites. No matter how careful a driver could be, these poor road conditions would still compromise one's ability to navigate safely. Moreover, severe crashes at certain infrastructure such as crossings and traffic signals indicate the need for clearer warning signs or even a redesign of the road.

Looking at the significant weather factors, the government could integrate detailed live weather reports on road signages or a mobile application. This initiative aids in making drivers more alert with their surroundings and potentially choose alternate, safer routes to get to their destination.

Lastly, the results obtained from the random forest algorithm shed light on the idea of using machine learning to support decision making. First responders, governments, and relevant agencies could deploy (or improve) the model above to identify more effective strategies to curb severe or minor crashes alike.

Conclusion

In this research, my main objective was to distinguish between minor and severe accidents with a machine learning model by looking at various temporal, geographical, and weather features. I first cleaned the dataset and then performed exploratory data analysis to build my understanding of possible patterns. With the insights I gained from this step, I constructed two machine learning models, a logistic regression with bi-directional AIC-based stepwise selection and a random forest. While both algorithms performed moderately well, I proceeded my analysis with the random forest as it yielded better scores. I discovered that these three broad factors do impact accident severity to some extent. My findings could be used by the responsible bodies to improve road conditions, provide drivers with safer alternative routes, and reevaluate existing crash prevention strategies. Looking ahead, it would be captivating to include human-behavior factors to prevent possible omitted variable bias.